Smaller Diophantine presentations of ZFC consistency

A synthesis of compact coefficients, separating norms, and the 31-variable quartic

AI-assisted research report for Vladimir Reshetnikov
ProveIt repository

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Comparison snapshot: `8c6eec9d19ef`

Abstract

We compare the seven audited wave-4 packages in ProveIt's collection and combine their complementary constructions. The largest improvement is a new complete fixed-coefficient encoding of the same 943-tile blank computation language, with $B = 2^{1,012,115}$ instead of $2^{1,953,125}$. Its four materialized integer ports total 506,664 bytes instead of 977,220, a 48.15% reduction. Port-specific label widths and a shared synchronization source marker are combined with mixed-radix clauses and an explicit prime-padding controller. Independently, exact algebra lowers the 12-positive-variable degree-1081 polynomial from 111 to 109 binary operations, and its three-equation system from 107 to 103. The degree-1069 alternative drops from 123 to 120 operations. The 31-variable quartic is combined with constant folding and a factored coefficient evaluator to reduce its nine-variable presentation from 1840 to 1587 operations over $\mathbb{N}$, or 1582 over $\mathbb{Z}_{>0}$, preserving its degree. All recommendations are concrete integer polynomials or systems with finite, halting-independent coefficient recipes. Composing the supplied reductions with the inherited compiler and calibrated K215 theorem proves in ZFC that their lack of solutions is equivalent to Con(ZFC). No global optimality or new end-to-end proof-assistant certificate is claimed.

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1 What was improved, and what "simplest" means

The requested property is an internal provable equivalence, not a consistency proof. With $\mathrm{Ref}_{\mathrm{ZFC}}$ denoting the usual arithmetized assertion that a finite ZFC refutation exists, our target is

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}, \qquad D = \mathbb{N} \text{ or } \mathbb{Z}_{>0}.$$

Negation gives exactly the user's requested statement. The domains are part of each construction: $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$ throughout. None of our positive-domain equations is silently interpreted over all of $\mathbb{Z}$.

There is no established single minimum simultaneously in variable count, degree, arithmetic evaluation size, and coefficient height. A nine-variable equation can have astronomical degree; a 67-operation expression has more variables but a far shorter evaluator. Accordingly, the report gives an explicit small collection of choices, with precise improvements on the current repository rather than an unsupported assertion of an absolute optimum.

1.1 Frozen comparison

We inspected the collection and its seven directories, numbered 24–30, at the following commit:

`8c6eec9d19ef80538c8e61ff6e0180040583446a`

The collection's current README reconciles prior packages and records the audit corrections; a byte copy is included in `evidence/COMPARISON_AT_PIN.md`. This avoids comparing against a superseded wave or a moving branch. The arithmetic and coefficient inputs used below have separate source manifests and hashes.

Operations mean binary $+$, $-$, and $\times$, including multiplication by small numerals or fixed coefficient ports. Reusing a value, referring to an integer leaf, and comparing two computed values are free. Fixed-coefficient arithmetic performed once during construction is not witness evaluation. Every supplied witness and every reported arithmetic gate is live.

Presentation	Variables	Degree	Old ops	New ops
Short positive polynomial	12	1081	111	**109**
Short positive system	12	1072	107	**103**
Tight positive polynomial	12	1069	123	**120**
Tight positive system	12	1060	118	**114**
Degree-best Jones form over $\mathbb{N}$	9	$120724\,5^{32} + 366$	1840	**1587**
Same Jones form over $\mathbb{Z}_{>0}$	9	$120724\,5^{32} + 366$	–	**1582**
Inherited product, new ports	15	155	67	67
Inherited system, new ports	16	24	59	59

For a system, the degree column is the largest residual degree. The short and tight systems have three equations; the inherited system has seven. The nine-variable degree is exactly

$$120724\,5^{32} + 366 = 2,810,824,662,446,975,708,007,812,866.$$

The new nine-variable evaluator does not beat every existing nine-variable evaluator in operation count: package 27 has smaller evaluators for different, higher-degree polynomials. It improves the evaluation of package 30's degree-best construction. Likewise, 109 does not beat the 67-operation 15-variable equation. These distinctions are essential to an honest comparison.

1.2 The seven packages contribute different ingredients

Package	Useful contribution	Role in this synthesis
24	Degree-1069 separating weight; arbitrary odd coefficient dimensions with an explicit prime controller; 24-integer form	Supplies the low-degree weight and the completeness controller for the smaller radix.
25	Mixed selector-count/payload radix; compact clause positions	Supplies the compact finite layout used by the new coefficient tuple.
26	Ratio-based roots and shifted transport	Confirms that transport normalization and norm choice are separate optimizations.
27	Fast nine-variable arithmetic; small coefficient instance; graph lifts at 59 operations	Supplies useful constant-folding and coefficient-evaluation ideas, and the detailed native-interface comparison.
28	Compact norm factorizations and structured Jones evaluator	Provides cross-checks and carefully separated inherited Pell dependencies.
29	Signed Jones construction with 17 unrestricted integers; signed finalizers	Establishes a distinct domain frontier that our positive constructions do not supersede.
30	31-variable quartic; trace/product norm evaluator; degree-best nine-variable specialization	Supplies the main positive template and the smaller quartic source for the new nine-variable evaluator.

The retained frontier also includes a positive quartic in 31 variables, several low-degree integer lifts, and the 17-integer Jones form. We have not improved those variable counts here. The current literature also contains Sun's ten-integer representation theorem [8]; that theorem is relevant to future integer-domain work but is not a ten-integer numerical instance supplied by this archive. Instantiating its full source-coding construction is a separate task. A general representation theorem must not be counted as a completed fixed-coefficient construction.

2 The logical chain and its verification boundary

The common semantic chain is

$$\begin{aligned}\text{polynomial/system has a solution} &\longleftrightarrow \text{marked periodic Wang943 tableau exists}\\ &\longleftrightarrow \text{literal K215 halts on blank tape}\\ &\longleftrightarrow \text{a finite ZFC refutation exists.}\end{aligned}$$

The first arrow includes the Pell and population-count compiler, not only a finite tile lookup. The final arrow has an inspected formal endpoint in `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint215.lean`, namely `blank_halts_iff_zfc_refutation`. Its stated theorem relates `BlankHalts machine215` to the syntactic bounded-proof predicate for ZFC refutation. A provenance copy is included in `proofs/`.

The present work did not rebuild the complete Lean or Rocq developments. Nor does it use the unqualified soundness of Lean as a theorem inside ZFC. The internal equivalence is obtained by the ordinary arithmetical proofs, which are formalizable in ZFC; the existing formal endpoint is corroborating and reusable mathematical infrastructure. The new norm identities and coefficient adapter are ordinary proofs plus finite certificates, not a new kernel-checked end-to-end development.

Theorem 2.1 (Combined consistency presentation)**.** *Bind each recommended arithmetic circuit to the complete coefficient tuple of Section 4. For the nine-variable circuits, also use their reconstructed quartic index and all fixed coefficient recipes, rather than those of the previous tuple. Then ZFC proves that the circuit's output has no zero in its stated domain, or its displayed system has no simultaneous solution, if and only if* Con(ZFC)*.*

Proof. The new coefficient adapter proves the same soundness and canonical completeness interface as the inherited blank compiler. The identities of Section 3 prove exact equality of the positive polynomial templates, and the shifted transport equations preserve existence of positive solutions. The inherited separating-root arguments therefore apply to those circuits. The small-operation baseline circuits use the unchanged inherited interface directly. For the nine-variable circuit, the retained quartic has the same zero-existence predicate, the coefficient bounds define a valid Jones index, and the new evaluator computes the same parametric Jones polynomial under those fixed leaves. Jones's reduction supplies the remaining equivalence. Composing these arithmetic equivalences with the calibrated machine theorem and negating gives the assertion. No choice in the coefficient construction depends on whether the machine halts. □

If ZFC is consistent, its second incompleteness theorem prevents ZFC from proving the displayed no-solution assertion. We make no unconditional two-sided independence claim: ruling out an erroneous existential proof requires an appropriate soundness assumption, not consistency alone. This logical distinction does not weaken the requested provable equivalence.

3 A twelve-variable polynomial with a shorter evaluator

3.1 Complete definition

All twelve supplied variables are strictly positive integers:

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t.$$

All symbols introduced next are polynomial abbreviations, not extra variables. Let β, K, M_C, M_F be the four fixed integers of Section 4, with M_F already shifted. Define

$$\begin{aligned}
q &= \beta J + 1, & X &= wq, & Y &= sq^3, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, & H &= 4a + 3,\\
\Delta &= a^2 + H, & \ell &= EkY, & A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2,\\
b_0 &= X + ac, & C &= q - F - \alpha.
\end{aligned}$$

Next put

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_k &= k - hE - R,\\
N_t &= (K + w)C + q - F - z(q-1),\\
I &= ic^2, \qquad D = 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \qquad d_0 = Q - 1,\\
b &= c + RD, \qquad P = QDc^2T^2, \qquad U = Qb^2.
\end{aligned}$$

Here d_0 is unrelated to the fixed coefficient exponent d below. For the short evaluator choose

$$M = k(I^2 + T^2), \qquad L = 1 + HM, \qquad W = b_0 + Ht.$$

The following is an entirely division-free evaluator:

$$\begin{aligned}
n &= (HM)^2 A, & D_n &= d_0 n,\\
H_P &= H^2 P, & H_U &= H^2 U,\\
H_e &= H_P + H_U - H^2, & j_0 &= 4 H_P H_U,\\
\mathcal{T} &= 2d_0(W^2 + L^2 G), & K_r &= d_0(W^2 - L^2 G), \qquad \mathcal{K} = K_r^2,\\
A_A &= 2(D_n + H_e), & c_0 &= (D_n - H_e)^2 - j_0,\\
a_t &= \mathcal{T} - A_A, & b_t &= c_0 - \mathcal{K},\\
c_1 &= 16 D_n j_0, & v &= c_1 - a_t(a_t \mathcal{T} + 2b_t),\\
u &= b_t^2 - a_t^2 \mathcal{K}, & N &= \mathcal{K} v^2 - u(\mathcal{T} v - u).
\end{aligned}$$

The recommended one-equation form is

$$\boxed{P_{109} = N_k N_t (1 + N) - 1 = 0.} \tag{1}$$

The literal binary schedule, including shared intermediates and operations by 2, 4, and 16, is `positive/circuits/short109_product.json`. Its exact total degree is 1081 and its evaluator has 109 operations.

For the system, use a shifted positive transport coordinate z, corresponding to $z_{\mathrm{old}} - 1$ in (1). The recommended three equations are

$$\boxed{\begin{aligned}
k - hE - R &= 1,\\
(K + w)C &= F + z(q - 1),\\
\mathcal{K} v^2 &= u(\mathcal{T} v - u).
\end{aligned}} \tag{2}$$

The supplied variables are still exactly the twelve listed above. This system has 103 operations and largest residual degree 1072. Replacing a zero-output test by a comparison is accounted for explicitly; it is not an uncharged subtraction concealed in the ledger.

3.2 Why the shorter norm is exactly the previous polynomial

For a formal argument z_0 define

$$\mathcal{F}(z_0) = \left[(d_0 z_0^2)^2 - A_A(d_0 z_0^2) + c_0\right]^2 - c_1(d_0 z_0^2).$$

If $g^2 = G$, the two expressions $x_\pm = d_0(W \pm Lg)^2$ satisfy

$$x_+ + x_- = \mathcal{T}, \qquad x_+ x_- = \mathcal{K}.$$

Reduce modulo the monic quadratic $x^2 - \mathcal{T}x + \mathcal{K}$:

$$\begin{aligned}
x^2 - A_A x + c_0 &\equiv a_t x + b_t,\\
(a_t x + b_t)^2 - c_1 x &\equiv u - v x.
\end{aligned}$$

Consequently

$$\begin{aligned}
\mathcal{F}(W + Lg)\mathcal{F}(W - Lg) &= (u - v x_+)(u - v x_-)\\
&= u^2 - \mathcal{T} u v + \mathcal{K} v^2\\
&= \mathcal{K} v^2 - u(\mathcal{T} v - u) = N.
\end{aligned}$$

This is an identity in integer polynomial rings. In particular, it remains valid when radicands vanish, become squares, or coincide. No division or generic nonsingularity assumption is involved.

To identify the inner expression with the inherited separating norm, put $e = P + U - 1$, $B_a = 2Q(cT)b$, and $m = HM$. The identity $j_0 = H^4 D B_a^2$ follows directly. Taking the product for $\tau = \pm\sqrt{A}$ in

$$[d_0(z_0 - m\tau)^2 - H^2 e]^2 - H^4 D B_a^2$$

produces $\mathcal{F}(z_0)$. Taking the remaining product for $g = \pm\sqrt{G}$ is exactly the preceding calculation. Thus the shorter evaluator computes the original complete norm, rather than an approximation or a polynomial merely agreeing at tested points.

There are two elementary savings. Computing

$$\mathcal{K} = [d_0(W^2 - L^2 G)]^2$$

uses one fewer multiplication than separately computing d_0^2 and the other square. Reversing the sign of the linear remainder coefficient costs nothing, and the final form $\mathcal{K}v^2 - u(\mathcal{T}v - u)$ uses six gates rather than the seven gates of the expanded three-term norm. In a system its final subtraction is unnecessary, saving one more gate. Finally,

$$\big[(K + w)C + q - F - (z + 1)(q - 1)\big] - 1 = (K + w)C - F - z(q - 1)$$

gives the transport saving. The inherited compiler bounds imply $C > 0$, $w > q > F$ at a solution, and hence $z_{\mathrm{old}} \geq 2$. The shifted coordinate therefore stays strictly positive in the reverse direction.

3.3 The semantic root argument is preserved

The arithmetic refactoring does not alter the delicate inverse argument. For orientation, the inherited short weight has the two strict separations

$$2(cT\sqrt{D} + b) < MkY^2 < \min\{M\sqrt{A}, M(\sqrt{G} - \sqrt{A})\}.$$

The outer transport and index equations supply the size hypotheses before any missing roots are recovered. Every complex root of the inner polynomial lies in the disk $|x| < M\sqrt{G}$. If G were nonsquare and the outer norm vanished, its two quadratic conjugates would give roots farther apart than that disk allows, because $L = HM + 1$. Thus G is an integer square. The same separation selects the positive main root, and rational recovery plus Pell saturation recovers the remaining integer witnesses.

For the product equation, unit signs also matter. The norm satisfies $N \equiv \square \pmod 4$, so $1 + N$ cannot be -1. A zero of (1) therefore forces $N = 0$ and gives the two outer factors a common sign. The inherited index argument removes the remaining negative case. This proof retains the initially possible $C = 0$ boundary until the signs have been settled. The archive includes the full predecessor proof, not just this outline, in `positive/upstream/POSITIVE12_PROOF.original30.md`.

3.4 The degree-1069 alternative

Keep every definition above, but replace the separating weight and carrier by

$$C_\sharp = E + 2(X + Y) + sq^2, \qquad M = I(2T + RC_\sharp ic), \qquad L = 1 + 2HM, \qquad W = Lb_0 + Ht.$$

The same trace/product evaluator now computes package 24's deliberately cancelling norm. It has 120 operations for the degree-1069 product and 114 for the shifted three-equation system, whose maximum

degree is 1060. Its existence theorem is inherited by exact polynomial identity and the same transport substitution.

The exact nonzero leading homogeneous form of this norm is

$$9\,2^{48} i^{48} (\eta+\zeta)^{96} w^{76} s^{176} (\beta J)^{648} (\beta J - F)^{16},$$

of degree 1060. The outer leading degrees are 7 and 2, giving 1069. For the short norm the leading term is $H_{e,\mathrm{top}}^{8}$, of degree 1072; the other expanded terms have smaller degrees. Its product therefore has exact degree 1081. These arguments hold uniformly for every authentic $\beta > 0$, rather than relying on a generic syntactic degree bound.

4 A smaller complete coefficient encoding

4.1 Finite language and fixed numerical parameters

The tile language is the same ordered list of 943 Wang tiles as in the source packages. It represents the literal blank-tape computation of K215, performs its actual first transition during initialization, and has distinguished Start and impossible Forbidden tiles at selector positions 0 and 1. The list is reconstructed from the 430 literal transitions and compared with its fixed SHA-256. Horizontal labels range up to 249; vertical labels range up to 438. Hence horizontal ports need eight identifier bits plus occupancy, whereas vertical ports need nine identifier bits plus occupancy.

Choose

$$b_c = 115, \quad L_c = 8801, \quad d = b_c L_c = 1,012,115, \quad \rho = 2^{115}, \quad B = \rho^{8801} = 2^d.$$

The four ports, always bound together, are

$$\boxed{\beta = B - 1, \qquad K = D_C + B D_R, \qquad M_C, \qquad M_F = M_{F,\mathrm{native}} + B - 1.} \tag{3}$$

These are specific integers. The following bounded sums give their full definition; the archive also contains their actual bytes.

Let the four ports have widths $r = (9, 10, 9, 10)$, starting field indices $f = (0, 9, 19, 28)$, and native copy offsets $o = (1023, 1013, 1003, 983)$. For port s and bit $0 \le j < r_s$, let $e_s(j) = o_s + j$. Use the occupied code $x + 2^{r_s - 1}$ for an edge label x on that port. Set

$$E_c = \{0, \dots, 942\} \cup \{e_s(j) : s < 4,\ j < r_s\} \cup \{1063, 1064, 1095, 2195, 3296\}.$$

Thus $|E_c| = 986$. Unlike the earlier layout, the source marker 1095 supplies both tested synchronization anchors, and marker 1096 is absent.

Put $w_j = 2^{10+2j}$ for $0 \le j < 41$. For tile i with edge labels $x_{i,s}$, define

$$c_i = 1 + \sum_{s<4} \sum_{j<r_s} \mathrm{bit}_j(x_{i,s} + 2^{r_s - 1}) w_{f_s + j} + w_{38} + w_{39} + w_{40}.$$

At each copy position set $c_{e_s(j)} = w_{f_s+j}$. The three anchors 1095, 2195, 3296 receive w_{38}, w_{39}, w_{40} respectively; the two dummy positions 1063, 1064 receive coefficient zero. Define

$$\mu = 1022 + \sum_{j=0}^{40} w_j, \qquad Z_0 = (2^{113} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{73} - 1)\rho^{8799}.$$

The literal coefficient recipe is

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e \in E_c} c_e(\rho^{4393-e} + \rho^{5495-e}),$$
$$D_R = \rho^{42},$$
$$M_C = B - 1 - \sum_{e \in E_c \setminus \{1\}} \rho^e - 2\rho^{1063},$$
$$M_{F,\text{native}} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{8} \rho^{1065+j} + \rho^{2195} + \rho^{3296} + Z_0.$$

All exponents here are fixed nonnegative integers. The bit-extraction symbols occur only in the finite construction of integer coefficients, not in the Diophantine equation evaluated on unknowns.

4.2 Decoding before assuming payload carries are absent

Suppose n selector bits are set in a native cell. Before decoding, $0 \le n \le 943$. Every other clause contribution is a multiple of 1024. The low ten binary bits of the clause sum therefore equal n. The low mask 1022 forbids all but bit 0, forcing $n = 0$ or $n = 1$.

Only after this conclusion do we examine the radix-four payload fields. Each is a sum of two Boolean bits, hence is in $\{0, 1, 2\}$ and cannot carry into its next field. Testing the low bit permits exactly 0 or 2. Consequently an occupied cell has the selected tile's edge bits and all three anchor bits; an empty cell has none. Like edge ports use the same width, so horizontal and vertical matching remain injective. The unequal widths introduce no comparison of incompatible codes.

The native overlap identities are

Target	D_C source	D_R source	Temporal source
$1013 + j$, $0 \le j < 10$	$983 + j$	none	$1013 + j$
$1065 + j$, $0 \le j < 9$	$1003 + j$	$1023 + j$	none
2195	1095	none	2195
3296	1095	none	3296

The new last row uses the bare shift 2201. Every scalar coefficient of these identities, and both complete clause identities, is checked on every basis position in E_c: 66,062 exact checks. By linearity they prove the stated finite identities for arbitrary native assignments.

4.3 Rotation synchronization, zero fields, and population

The exact circular support checks are

$$((4393 - E_c) \bmod 8801) \cap ((5495 - E_c) \bmod 8801) = \varnothing,$$
$$((2195 - E_c) \bmod 8801) \cap ((3296 - E_c) \bmod 8801) = \{0\}.$$

For an explanatory integer calculation, 1102 is absent from $E_c - E_c$, and 1101 occurs only as $3296 - 2195$; moreover $8801 > 2 \max E_c$ excludes an additional circular case. These are the two clean-band and unique-anchor conditions used by the native compiler. Synchronization starts at the marked Start cell: selector zero is already one there, and a clean clause forces its shared source anchor to one. The two tested anchors remain distinct even though their source is shared. The binary parity argument therefore still rules out a nonzero bit alignment, including the single upper-dummy spill exception.

Whole-cell shifts then recover the same Wang matching semantics; only then does spatial matching propagate occupation globally. No coprimality of a decoded temporal shift is assumed.

The conservative carry mass is

$$\mathcal{M} = (986 + 2)\left(2\sum_{e\in E_c} c_e + 6\right).$$

Its bit length is 112, and $4\mathcal{M} + 8$ has bit length 114. The maximum zero-field mask digit is $2^{113} - 1$, with $2(2^{113} - 1) + 4 < 2^{115}$. The same bounds control carries before decoding and preserve the strict canonical slack.

The largest unshifted computation exponent is $5495 + 3296 = 8791$. All 864 padding bits in Z_0 are above this exponent, in fields 8792–8799, and are identically zero on aligned computations, including safe dummy changes. They strengthen soundness without excluding a canonical accepting tableau. Their number is $7 \cdot 113 + 73 = 864$. Since $\mathrm{pc}(\mu) = 50$, the high-mask population is

$$1 + 2 \cdot 50 + 10 + 9 + 2 + 864 = 986.$$

The exact materialized integers satisfy

$$\mathrm{pc}(M_C) = d - 986, \quad \mathrm{pc}(M_{F,\mathrm{native}}) = 986, \quad M_C \equiv 2 \pmod 4, \quad M_{F,\mathrm{native}} \equiv 4 \pmod 8,$$

as well as $0 < M_C, M_{F,\mathrm{native}} < B - 1$. The population identity required by the Pell compiler is therefore unchanged.

4.4 Completeness with a new explicit prime controller

Shrinking the radix destroys the old power-of-five argument; checking just the masks would not establish completeness. Here

$$d = 5 \cdot 13 \cdot 23 \cdot 677, \qquad \lambda = 3433, \qquad k_c = 3432 = \lambda - 1.$$

Trial division verifies primality of λ. Exact modular exponentiation gives

Modulus	5	13	23	677	3433
Order of B	4	12	11	52	132

Thus $\mathrm{ord}_d(B) = 1716$ divides k_c. Furthermore

$$B^{k_c} \bmod \lambda^2 = 1,973,976 \neq 1, \qquad v_\lambda(B^{k_c} - 1) = 1.$$

For $H_0 = 1 + BD_C + B^2(D_R + 1)$ the unit certificate is

p	$B \bmod p$	$(B^2 - 1) \bmod p$	$H_0 \bmod p$
5	3	3	2
13	7	9	12
23	9	11	20
677	665	143	602
3433	3241	2533	3121

Choose sufficiently large canonical strip width w_c and height h_c, both powers of λ, using independent rectangular padding. Put $N_c = w_c h_c = \lambda^n$, $q = B^{N_c}$, and ensure $N_c > d\lambda^2$ and $h_c > \lambda$. All relevant orders divide $\lambda - 1$, so $q \equiv B^{w_c} \equiv B \pmod{d\lambda}$. Switching the low dummy bit in cell j changes the actual packed index by

$$\delta R = -\Gamma B^j, \qquad \Gamma = \rho^{1063}(q^2 - 1)[1 + q(D_C + BD_R + B^{w_c})].$$

The preceding table proves $\gcd(\Gamma, dN_c) = 1$.

Lemma 4.1 (Boolean coverage of the actual modulus)**.** *Let $T_c = N_c/\lambda$. Every residue modulo dN_c is a Boolean subset sum of*

$$\{B^{2+k_c j} : 0 \le j < T_c\} \cup \{B^3\}.$$

The associated dummy cells and all required spatial and temporal translates can be chosen without wrapping.

Proof. Set $C_0 = d\lambda$, so $dN_c = C_0 T_c$ and $T_c > C_0$. The valuation certificate and elementary binomial lifting show that B^{k_c} has order T_c modulo λ^n and is 1 modulo d. Its first T_c powers are exactly the residues $1 + C_0 z$ modulo $C_0 T_c$.

Normalize a desired residue by B^{-2}. Choose $\varepsilon \in \{0, 1\}$ so that subtracting εB leaves a nonzero residue modulo λ. Let a be the least residue of this difference modulo C_0. Then $1 \le a < C_0 < T_c$ and $\gcd(a, T_c) = 1$. Write the remaining target as $a + C_0 u$, and solve

$$az_0 + \frac{a(a-1)}{2} \equiv u \pmod{T_c}.$$

The a distinct orbit entries with additive coordinates $z_0, z_0 + 1, \ldots, z_0 + a - 1$ modulo T_c have the desired sum; include B^3 exactly when $\varepsilon = 1$.

The largest selected position is at most $N_c - N_c/\lambda - \lambda + 3$. Since $h_c > \lambda$, one has $w_c < N_c/\lambda$, so translations by 1 and w_c stay below N_c. Position 3 is distinct and satisfies the same bounds. No cell is reused. □

Starting from a canonical halting tableau with initial packed index R_0, apply the lemma to $(R_0 - dw_c)\Gamma^{-1} \pmod{dN_c}$. The modified tableau satisfies $R \equiv dw_c \pmod{dN_c}$, precisely the temporal rotation required by $X = 2^R$. These safe dummy changes preserve all tile, marker, overlap, mask, carry, and strict slack conditions. The inherited Pell converse then supplies the remaining positive arithmetic witnesses.

All dimensions d, N_c are odd. The original population and low-mask identities still give $R \equiv 3 \pmod 4$ and $\mathrm{pc}(R) = 3dN_c + 2$. Consequently the canonical power scale is a square, and the canonical strong multiplier and s-scale can be chosen odd by the same source argument. This proves the full adapter, including the extra canonical promises needed by the integer-domain source families, rather than merely the congruences of four arbitrary integers.

4.5 Materialized coefficient sizes

Port	Bits	Raw bytes	Gzip bytes
β	1,012,115	126,515	159
K	1,016,946	127,119	9,735
M_C	1,012,115	126,515	262
M_F (shifted)	1,012,116	126,515	260
Total		506,664	10,416

These are unsigned minimal-width little-endian payloads. The new total is 48.15% below 977,220 bytes, while the largest port's bit length falls 48.07%. This measures the four fixed program ports, not the largest coefficient after expanding an entire high-degree polynomial. Compression size is also not coefficient height. The source tuple of package 25 or 27 is replaced as a whole; changing just β would be invalid.

5 Combining the 31-variable quartic with the faster Jones evaluator

5.1 The parent quartic is explicit

Package 30 eliminates one packing register from the earlier 32-variable quartic. Its 31 positive coordinates are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta.$$

With affine abbreviations $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, and $C = q - F - \alpha$, the 22 equations are

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L+1), & X &= wq,\\
a &= E + sN, & E &= X(a - E), & c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1,\\
D &= X + ac + \gamma H, & \Delta &= a^2 + H, & C_2 &= c^2,\\
D^2 &= \Delta C_2 + 1, & k &= jc - V + 1 + hE.
\end{aligned}$$

$$jc - V = (q - F)(N - q) - CL + (M_C + qM_F)J,$$
$$(K + w)C + (q - F) - z(q - 1) = 1.$$

$$\begin{aligned}
F_2 &= f^2, & V &= of - c, & g &= y^2 - V^2,\\
I &= iC_2, & B_a &= I^2, & B_a &= \Delta(F_2 - 1),\\
y^2 - B_a g &= 1.
\end{aligned}$$

Let Q_{31} be the sum of the squares of their left-minus-right residuals. Every residual is quadratic, so Q_{31} is quartic. The coefficient of τ^4 is 1, proving exact degree four for every valid fixed tuple. At the positive all-one tuple the residual $E - X(a - E)$ is 1; hence shifting every positive coordinate by $+1$ gives a quartic over $\mathbb{N}^{31}$ with strictly positive constant term, as required by the Jones normalization. The complete sparse polynomial is retained in `nine/source/quartic31.json`.

The one-register elimination is an exact positive zero-set bijection. The omitted register was $G = q(q - F) - C$, and the identity

$$L[q(q - F) - C] - [(q - F)(N - q) - CL] = (q - F)[q(L + 1) - N]$$

restores the packing equation. Transport forces $C > 0$, and then $G = (q - 1)C + q\alpha > 0$. We did not find a justified second elimination that preserves quadratic residuals. It would be incorrect to report a 30-variable quartic merely by dropping a positivity obligation or raising the residual degree.

5.2 All Jones index constants are specified

Introduce one unused input coordinate, fixed to 1, and let p_α be the coefficients of the shifted quartic, with $\alpha \in \mathbb{N}^{32}$ and $|\alpha| \le 4$. Write

$$c_\alpha = (4 - |\alpha|)! \left(\prod_{j=0}^{31} \alpha_j! \right) p_\alpha, \qquad e(\alpha) = \sum_{j=0}^{31} \alpha_j 5^j, \qquad L_J = 5^{32}.$$

There are 402 nonzero coefficient records; each is a specified polynomial in the four fixed compiler ports. Every omitted record is explicitly zero.

Let the port bit lengths, in the order β, K, M_C, M_F, be

$$(b_1, b_2, b_3, b_4) = (1012115, 1016946, 1012115, 1012116).$$

For a record $c_\alpha = \sum_t a_t \prod_r \mathrm{port}_r^{e_{tr}}$, define the exact integer upper bound

$$H_\alpha = \sum_t |a_t| 2^{\sum_r b_r e_{tr}}.$$

All records satisfy $|c_\alpha| \le H_\alpha$, and exact finite arithmetic checks $H_\alpha + 1 < 2^{4{,}058{,}127}$. Therefore set

$$\boxed{z_J = 2^{4{,}058{,}127}.} \tag{4}$$

The other essential fixed index is

$$y_J = \sum_{\substack{\alpha \in \mathbb{N}^{32} \\ |\alpha| \le 4}} (z_J + c_\alpha)(2z_J)^{L_J - e(\alpha)}. \tag{5}$$

This is a bounded sum of exactly $\binom{36}{4} = 58{,}905$ terms. Its digits are between 1 and $2z_J - 1$, and their positions are distinct. Both y_J and z_J are fixed integers, not unknowns, oracles, or unspecified enumeration indices. A conservative binary-length bound for y_J is $4{,}058{,}128(5^{32} + 1)$. Its enormous size is a genuine cost of the nine-variable presentation and is not hidden by the short recipe.

The actual z_J is supplied in compressed binary form: 507,266 raw bytes, 527 gzip bytes. The previous instance used $z_J = 2^{45{,}933{,}757}$. The new bound is recomputed from the new tuple. Carrying over the old coefficient records numerically while changing only the four compiler ports would not be a coherent instance.

5.3 The evaluation identity

The expensive fixed digit polynomial in a variable base $\mathcal{B}$ is

$$e_J(\mathcal{B}) = \sum_{|\alpha| \le 4} (z_J + c_\alpha) \mathcal{B}^{L_J - e(\alpha)}.$$

Set $T_J = 5^{31}$ and introduce computed registers

$$t_j = \mathcal{B}^{5^j} \quad (0 \le j \le 32), \qquad v_j = \mathcal{B}^{T_J - 5^j} \quad (0 \le j \le 31).$$

These are fixed-exponent polynomials; none is an extra witness. They are computed through $t_{j+1} = t_j^4 t_j$, $v_{31} = 1$, and $v_j = t_j^4 v_{j+1}$. The elementary exponent identity

$$\mathcal{B}^{L_J - e(\alpha)} = t_{31}\, t_{31}^{4 - |\alpha|} \prod_j v_j^{\alpha_j}$$

has no negative exponent or division. If h_4 denotes the complete homogeneous polynomial of degree four, it gives

$$e_J(\mathcal{B}) = t_{31}\left[z_J h_4(t_{31}, v_0, \ldots, v_{31}) + \sum_{\alpha} c_\alpha t_{31}^{4-|\alpha|} \prod_j v_j^{\alpha_j}\right]. \tag{6}$$

The complete homogeneous term includes every zero coefficient position automatically. It is evaluated by the usual ascending-order updates $h_r \leftarrow h_r + w h_{r-1}$ for $r = 1, \ldots, 4$, initialized with $h_0 = 1$ and the other registers zero. The sparse correction is evaluated by multivariate Horner factorization. Literal records become literals, and identical coefficient recipes share a fixed leaf. Both transformations are exact identities in the compiler ports. In total, the new DAG has only 80 live fixed leaves: z_J, y_J, and 78 distinct nonliteral recipes.

The geometric register also factors without division. With $q_J = \mathcal{B}^{L_J}$,

$$\sum_{r=0}^{4L_J-1} \mathcal{B}^r = \left[\prod_{j=0}^{31}(1 + t_j + t_j^2 + t_j^3 + t_j^4)\right](1 + q_J)(1 + q_J^2),$$

and each five-term factor equals $1 + t_j(1 + t_j)(1 + t_j^2)$.

All remaining Jones expressions and the source's tight two-root finalizer are retained verbatim as polynomial data. The complete executable definition, with exactly nine supplied inputs, is `nine/compact/nine31_natural.json`. The affine positive-coordinate version is `nine/compact/nine31_positive.json`. The finalizer has the form

$$[\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)]^2 - 4\Omega^4 V_0^2 a_0 a_1.$$

Its input expressions are all defined in those literal DAGs; the source proof bounds both radicands before imposing square conditions and treats negative or coincident radical cases. We preserve this finalizer rather than replacing its proof with generic claims about independent radicals.

5.4 Metrics, proof, and remaining cost

Exact expansion verifies the sparse correction in (6) as 675 matching symbolic terms over $\mathbb{Z}[\mathcal{B}, \beta, K, M_C, M_F]$. The recurrence and exponent identity prove its remaining baseline. This gives the exact original polynomial at the old binding, in 1587 operations instead of 1840: a 253-operation, or 13.75%, saving. The gate breakdown is 857 multiplications and 730 additions/subtractions. The source's leading-degree certificate is uniform in the index, with diagonal leading coefficient $2^{274} z_J^2 > 0$. Thus coherent rebinding preserves the exact degree as well.

Replacing each nonnegative supplied coordinate by a positive coordinate minus 1 cancels several internal shifts. The resulting evaluator costs 1582 operations and has the same degree and zero-existence condition. The transformation is an explicit affine bijection, not an informal domain change.

Jones's nine-variable reduction [6, 7], the normalized quartic, and the strict index bounds give the semantic theorem. The complete coefficient recipes and proof are in `nine/NINE_PROOF.md` and `nine/compact/CONSTANTS.json`. Although every coefficient is effectively specified, y_J is not materialized. The nine-variable form is consequently the coordinate-minimizing choice among the recommendations, not the smallest actual numerical object or the easiest equation for human review.

6 The shorter inherited presentations, with the new tuple

For users prioritizing a concise system, the seven-equation 59-operation presentation remains useful. Its sixteen positive variables are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma.$$

Use $q, X, Y, E, k, c, a, H, \Delta, C, R$ as defined in Section 3, but in this section put

$$D = X + ac + \gamma H, \quad L_0 = EkY, \quad V = of - c, \quad U_0 = jc - R, \quad I = ic^2, \quad A_0 = \Delta(f^2 - 1).$$

Then the complete system is

$$\begin{aligned} \tau^2 - L_0(L_0 + k) &= 1, & D^2 - \Delta c^2 &= 1, \\ A_0(V^2 - y^2) + y^2 &= 1, & k - hE - R &= 1, \\ (K + w)C &= F + z(q - 1), & I^2 &= A_0, \\ U_0 &= V. \end{aligned} \tag{7}$$

Here z is the shifted transport coordinate. The largest residual degree is 24. Graph lifts can lower this degree at the same arithmetic count but add witnesses and equations; those existing tradeoffs are not recounted as new results.

For one polynomial at the smallest inherited arithmetic count, retain the fifteen positive coordinates

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma.$$

Use the same common abbreviations and set

$$\begin{aligned} V &= c(Tf - 1) - Rf^2, & K_a &= (i\Delta c^2)^2, \\ N_0 &= \tau^2 - L_0(L_0 + k), & N_1 &= D^2 - \Delta c^2, \\ N_a &= K_a(V^2 - y^2) + y^2, & N_s &= \Delta f^2 - K_a, \\ N_k &= k - hE - R, & N_t &= (K + w)C + q - F - z(q - 1). \end{aligned}$$

Here z is the unshifted coordinate. The equation is

$$\boxed{N_0 N_1 N_a N_k N_t N_s - \Delta = 0.} \tag{8}$$

It has 67 operations and exact degree 155. Its complete inherited unit-sign and Pell proof is included in `proofs/INHERITED_P67_PROOF.md`; the new local contribution to this instance is the smaller complete coefficient tuple. We do not claim a 66-operation polynomial.

The strongest caution in this proof is useful more generally: weak transport initially permits $C = 0$ and a negative unit sign. Discarding that boundary before recovering the main Pell index would invalidate the argument. The source proofs retain the boundary, recover the target index, eliminate the negative sign, and only then conclude $C > 0$.

7 Verification, reproducibility, and provenance

7.1 What the finite checks establish

The archive is intended to be an additive, reviewable research package. It includes the actual four coefficients, source machine transitions, literal arithmetic circuits, exact coefficient recipes, source pins,

mathematical proofs, standard-library Python programs, LaTeX source, and the compiled PDF. No network request is required to replay the main checks.

Component	Verification and its scope
Finite language	Reconstruct all 943 ordered tiles from the 430 literal machine actions, and verify the complete tile-list hash.
Coefficient adapter	Independently reconstruct all four full integers; check 66,062 native-basis identities, exact populations and congruences, carry thresholds, support intersections, primality, orders, lifting, and actual unit residues.
Positive norms	Prove the quadratic-remainder identity algebraically; check exact eight-conjugate evaluations on signed fixtures, predecessor agreement, transport substitutions, liveness and operation ledgers, and full polynomial degree diagnostics in two finite fields.
Nine-variable evaluator	Check 675 exact symbolic correction terms, 12,032 intermediate-value comparisons, and 128 affine-coordinate checks; verify all supplied inputs/gates are live.
Jones constants	Verify actual compiler payload hashes and bit lengths, all 402 strict index bounds, the complete zero-record convention, and the actual z_J payload.
Release	Run the package from its delivered relative layout, retain receipts and hashes, compile the TeX document, and inspect rendered pages.

These checks do not test whether the actual consistency polynomial has a zero. They also do not replace the infinite-direction compiler or Pell arguments. Random agreement is corroboration, whereas a polynomial identity is proved by the displayed algebra or exact sparse comparison. Finite-field degree checks corroborate the uniform leading-term proofs; they are not the sole basis for the exact degree claims.

7.2 Commands and bindings

From the package root, run

```
python3 verify.py
python3 reproduce.py
```

The first performs the release checks. The second regenerates the new coefficients and circuits before checking them. Both use Python 3.10 or later with the standard library. To rebuild the article:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Every recommended circuit identifies its variable domain and coefficient binding. The names K_0 and `Kconstant` alias K; `Bm1` aliases β; `MF` aliases the shifted M_F. Historical files in `source/` or `upstream/` retain their original coefficients and are explicitly comparisons. The recommended nine-variable circuit has its own newly derived index constants. No old Jones index is silently paired with a different quartic.

The only use of the remote repository was read-only source inspection. No commit, branch, pull request, issue, or other remote object was created. Source license and attribution files are preserved. A proposed placement is a new directory under the user's existing `con-zfc-diophantine-forms` collection; the user can incorporate it using the repository's preferred intake process.

8 What remains worth trying

Several plausible shortcuts fail to establish what they appear to promise. Deleting $N_k = 1$ and its coordinate h would appear to give eleven positive variables. But that equation supplies $k \equiv R + 1 \pmod{E}$, hence the first-Pell-index congruence $2n \equiv R + 1 \pmod{E}$. After main-index recovery it forces the exact relation needed by the half-binomial population argument. The strict ratio alone gives bounds, not this congruence. No eleven-variable reduction is claimed.

Likewise, replacing β alone by a smaller base, replacing a main congruence-carrying root by its absolute value, or merging radical tests without checking repeated or negative radicands can introduce extraneous solutions. Coefficient numerical audits establish an interface only when accompanied by its completeness controller and semantic proof.

The next substantial gains are likely to require one of three more structural advances. A smaller quadratic-residual parent would reduce the $5^{\nu+1}$ exponent in the Jones construction. A new way to encode the first-index congruence could remove an actual moderate-degree witness. A different finite computation language or more compact native geometry could reduce the fixed ports beyond the conservative layout retained here. These are specific remaining proof obligations, not asserted results.

For incorporation now, the 59-operation system is the shortest transparent display, the 103-operation twelve-variable system is a useful balance of coordinates and evaluation size, and the 1582-operation nine-positive- variable polynomial is the coordinate-minimizing recommendation. Each has an explicit fixed instance and the requested ZFC-provable equivalence.

References

[1] Vladimir Reshetnikov and contributors. *Con(ZFC) in Diophantine form*, audited wave-4 collection. ProveIt, commit `8c6eec9d19ef80538c8e61ff6e0180040583446a`. Pinned collection. Local copy: `evidence/COMPARISON_AT_PIN.md`.

[2] ProveIt report packages 24 and 30. *Smaller explicit Diophantine forms of ZFC consistency*; *A smaller twelve-positive-variable synthesis*. Pinned source proofs and polynomial circuits are recorded in `positive/SOURCE_PINS.json`.

[3] ProveIt report packages 24, 25, and 27. Mixed-radix native compiler, Wang943 language, arbitrary-odd-dimension prime padding, and complete coefficient adapters. Exact source paths and SHA-256 values: `coefficients/SOURCES.json`; retained proofs: `coefficients/source_proofs/`.

[4] ProveIt report package 30. *A 31-positive-variable quartic by eliminating a packing register* and *The compact nine-variable construction*. Exact input manifest: `nine/SOURCE_MANIFEST.json`.

[5] ProveIt. *KEndpoint215.lean*. Pinned formal endpoint. Inspected, not rebuilt in this work; provenance copy in `proofs/`.

[6] James P. Jones. Universal Diophantine equation. *Journal of Symbolic Logic* 47(3), 549–571 (1982). doi:10.2307/2273588.

[7] ProveIt. Corrected Jones manuscript and parameterized nine-variable development. `Computability/HilbertTenthProblem/Papers/1982/` and `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1982/`. The present composition uses the ordinary source theorem as identified in the retained package-30 proof, not a newly rebuilt formal specialization.

[8] Zhi-Wei Sun. *Ten unknowns for Hilbert's tenth problem over the integers*. arXiv:2609.01594v2 (2 September 2026). Versioned primary source. This general theorem is discussed for comparison only and is not instantiated as a ten-integer deliverable here.